PES1UG24CS905
SAMARTH PRABHU K

CSE – D

1A.

```
root@Samarth:~/PES1UG24CS905-pes-vcs# git commit --allow-empty -m "Phase1: completed integrity verification and data extraction"
[main 95d08c5] Phase1: completed integrity verification and data extraction
 Committer: root <root@Samarth.localdomain>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

root@Samarth:~/PES1UG24CS905-pes-vcs# git push
```

1.B

```
root@Samarth:~/PES1UG24CS905-pes-vcs# find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
```

2.A

```
root@Samarth:~/PES1UG24CS905-pes-vcs# find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
```

2.B

```
All Phase 1 tests passed.
root@Samarth:~/PES1UG24CS905-pes-vcs# xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12 | head -20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
root@Samarth:~/PES1UG24CS905-pes-vcs# xxd .pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476 | head -20
00000000: 626c 6f62 2031 3800 4475 706c 6963 6174  blob 18.Duplicat
00000010: 6520 636f 6e74 656e 740a                 e content.
root@Samarth:~/PES1UG24CS905-pes-vcs# xxd .pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf | head -20
00000000: 626c 6f62 2031 3500 5465 7374 2069 6e74  blob 15.Test int
00000010: 6567 7269 5879 0a                        egriXy.
```

3.A

```
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
root@Samarth:~/PES1UG24CS905-pes-vcs# rm -rf .pes
./pes init
./pes add file.txt
./pes status
cat .pes/index
Initialized empty PES repository in .pes/
Added file(s) successfully
Staged changes:
  staged:     file.txt

Unstaged changes:

Untracked files:
  untracked:  pes.o
  untracked:  test_tree.c
  untracked:  object.c
  untracked:  test_sequence.sh
  untracked:  index.h
  untracked:  pes.h
  untracked:  tree.h
  untracked:  pes.c
  untracked:  tree.o
  untracked:  commit.h
  untracked:  test_objects.c
  untracked:  object.o
  untracked:  commit.o
  untracked:  README.md
  untracked:  pes
  untracked:  commit.c
  untracked:  index.c
  untracked:  Makefile
  untracked:  tree.c
  untracked:  index.o
100644 0bd69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d file.txt
```

3.B

```
root@Samarth:~/PES1UG24CS905-pes-vcs# cat .pes/index
100644 0bd69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d file.txt
```

4.A

```
root@Samarth:~/PES1UG24CS905-pes-vcs# ./pes commit -m "phase 4 commit"
error: commit failed
root@Samarth:~/PES1UG24CS905-pes-vcs# ./pes commit -m "first commit"
error: commit failed
root@Samarth:~/PES1UG24CS905-pes-vcs# nano commit.c
root@Samarth:~/PES1UG24CS905-pes-vcs# nano commit.c
root@Samarth:~/PES1UG24CS905-pes-vcs# make
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
root@Samarth:~/PES1UG24CS905-pes-vcs# rm -rf .pes
./pes init
./pes add file.txt
./pes commit -m "first commit"
Initialized empty PES repository in .pes/
Added file(s) successfully
Committed: 09e3b55ae179... first commit
root@Samarth:~/PES1UG24CS905-pes-vcs#
```

```
root@Samarth:~/PES1UG24CS905-pes-vcs# ./pes log
commit 170e11c1ae663ba330b397608bee48fa572c638b6ca7971eb8628554b16e31f9
Author: PES User <pes@localhost>
Date:   1776787527

    third commit

commit 6297d302e1d55c92c6be6f0aaf4ee7d5509bfb9baad0368788543430693a5a02
Author: PES User <pes@localhost>
Date:   1776787526

    second commit

commit 09e3b55ae1791e92a80961a8ff58b46e5daf473ec552ca686b805f22af17f6d0
Author: PES User <pes@localhost>
Date:   1776787429

    first commitirst commit<Q
```

4.B

```
root@Samarth:~/PES1UG24CS905-pes-vcs# find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/09/e3b55ae1791e92a80961a8ff58b46e5daf473ec552ca686b805f22af17f6d0
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/17/0e11c1ae663ba330b397608bee48fa572c638b6ca7971eb8628554b16e31f9
.pes/objects/19/f5043660c808b7ed1db6a44b8025f0d17788eebeb15c19cd73cf4b7b6dc4c9
.pes/objects/2a/9bdd8d919f86af7a26f7caea489fcb59ddd0be2a004712f686e7899c4937be
.pes/objects/5a/e6cd212190f6f970c2b30f000f1253a0d3daac7f491396f35f80cc30888a4e
.pes/objects/62/97d302e1d55c92c6be6f0aaf4ee7d5509bfb9baad0368788543430693a5a02
.pes/objects/97/abe04ed898f85a4c6f7059cce71df9c5f994bb7da187b251b7443ef6ed23d0
.pes/objects/9c/b22f8097895c13c57aeb33e805d440f9f9572ed29ab332a02fa8729e35fabc
.pes/refs/heads/main
```

4.C

```
root@Samarth:~/PES1UG24CS905-pes-vcs# cat .pes/refs/heads/main
cat .pes/HEAD
170e11c1ae663ba330b397608bee48fa572c638b6ca7971eb8628554b16e31f9
ref: refs/heads/main
```

root@Samarth:~/PES1UG24CS905-pes-vcs# make test-integration
= Running integration tests =

```
bash test_sequence.sh
= PES-VCS Integration Test =

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Added file(s) successfully
Status after add:
Staged changes:
staged: file.txt
staged: hello.txt

Unstaged changes:

Untracked files:

--- First Commit ---
Committed: 8d442ed87ce9... Initial commit

Log after first commit:
commit 8d442ed87ce92af985391cb7893a81d88686e13e34fbe3189ab5593033d38b3b
Author: PES User pes@localhost
Date: 1776787869

Initial commit

--- Second Commit ---
Added file(s) successfully
Committed: 11e5bbd01d38... Update file.txt

--- Third Commit ---
Added file(s) successfully
Committed: 458b664feede... Add farewell

--- Full History ---
commit 458b664feede2679893cb060489d0be7eb112c70b351396a5287a9a22b1ce6d5
Author: PES User pes@localhost
Date: 1776787869

Add farewell

commit 11e5bbd01d382cfcd9d64f5f2ed20a7771b5174469f5dcd947847ddf6699b9a6
Author: PES User pes@localhost
Date: 1776787869

Update file.txt

commit 8d442ed87ce92af985391cb7893a81d88686e13e34fbe3189ab5593033d38b3b
Author: PES User pes@localhost
Date: 1776787869
```

Initial committial commiQ

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
458b664feede2679893cb060489d0be7eb112c70b351396a5287a9a22b1ce6d5

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/11/e5bbd01d382cfcd9d64f5f2ed20a7771b5174469f5dcd947847ddf6699b9a6
.pes/objects/45/8b664feede2679893cb060489d0be7eb112c70b351396a5287a9a22b1ce6d5
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/8d/442ed87ce92af985391cb7893a81d88686e13e34fbe3189ab5593033d38b3b
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

= All integration tests completed =